Supersonic Flow over a Double Ramp

Angewandte CFD — Project RB1

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Course Angewandte CFD (MW1628)
Tools ANSYS CFX ICEM CFD

Solver Settings Summary

Advection Scheme	High Resolution
Transient Scheme	2nd Order Backward Euler
Turbulence Numerics	High Resolution
Convergence Criterion	RMS, target 10^{-6}
Time Step Control	Adaptive (CFL = 0.8), $\Delta t_0 = 4.5 \times 10^{-5}$ s
Fluid	Air (ideal gas), $T_0 = 900$ K, $p_{\text{ref}} = 0$ atm
Mach Number	$Ma = 3.8$ ($u_x = 1159$ m/s)
Fluid Model	Total Energy (incl. viscous work)
Turbulence Models	Inviscid / k-ω / k-ε
Domain	60×40 m
Grid Size	286 995 nodes; 117 904 quads; 227 520 hexas
Grid Dimension	3D (thin layer in z)
Quality Criteria	Vol.> 0; AR 0.3–0.5; Det. ≈ 1; $\theta_{\min} = 68$

Boundary Conditions and Setup

Choice of Boundary Conditions

The simulation models a 2D supersonic airflow ($Ma = 3.8$) over a double-ramp configuration ($\alpha = 8$, $\beta = 22$). At this Mach number, density variations are significant, requiring a compressible formulation with isoentropic relations to determine the inlet state. Using $T_0 = 900$ K and $Ma = 3.8$, the static temperature is $T_s = 231.48$ K, the speed of sound $a \approx 305$ m/s, the inlet velocity $u_x = 1159$ m/s, and the static pressure $p_s = 869$ Pa.

- **Supersonic Inlet.** All flow variables are fixed (p_0, T_0, velocity) because information travels only downstream at $Ma > 1$.
- **Supersonic Outlet.** No back-pressure is imposed; disturbances cannot propagate upstream, so the outlet is left unconstrained.
- **Ramp — Free-Slip Wall** (inviscid run). Viscosity is absent; the fluid slides freely along the surface, allowing pure shock/expansion analysis.

- **Ramp — No-Slip Wall** (viscous runs). Enforces $\mathbf{u} = \mathbf{0}$ at the wall, enabling boundary-layer development and shock–boundary-layer interaction.
- **Farfield — Free-Slip Wall.** Placed 40 m from the ramp; viscous effects are negligible at this distance and no artificial reflections are introduced.
- **Lateral Faces — Symmetry.** ANSYS CFX requires 3D grids; symmetry planes enforce 2D behaviour in the thin extrusion direction.

Fluid Model and Turbulence Model

Air is treated as a calorically perfect ideal gas with the Total Energy heat-transfer model to capture the coupling between kinetic and thermal energy at high speed. Three simulations are performed:

- **Inviscid.** No turbulence model; used as a baseline to isolate compressibility effects (shock waves, Prandtl–Meyer expansion fans) without viscous contamination.
- **k-ω.** Well-suited for adverse-pressure-gradient boundary layers and shock-induced separation; superior near-wall accuracy but sensitive to free-stream ω specification.
- **k-ε.** Robust convergence for free-shear and external flows; less accurate in the inner boundary layer and near strong pressure gradients.

Discretisation Method

The *High Resolution* advection scheme blends between first- and second-order accuracy in a solution-dependent manner, minimising numerical diffusion while suppressing spurious oscillations near shocks. The Second-Order Backward Euler scheme is used for time advancement, providing second-order temporal accuracy for transient effects.

Grid

Mesh Refinement and Quality

The structured mesh is refined in two critical regions. Near the ramp walls, geometric progression is applied in the wall-normal direction to resolve the boundary layer and achieve target y^+ values compatible with the k-ω wall functions. In the shock-wave regions (both oblique shocks and the reflected shock), cells are clustered to reduce numerical diffusion and sharpen shock-wave capturing.

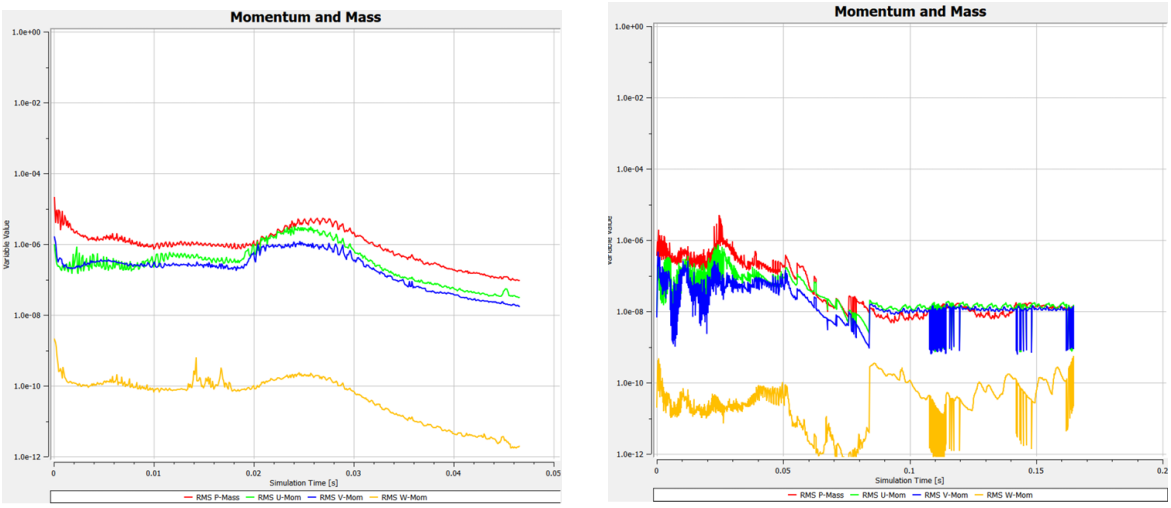
Grid quality is assessed with four criteria, all satisfied:

Table 1. Grid quality metrics.

Criterion	Value achieved	Requirement
Cell Volume	All > 0	> 0
Aspect Ratio	0.3–0.5	< 10
Determinant	≈ 1.0	> 0.3
Minimum Angle	68	> 18

The determinant close to unity confirms cells are free of geometric distortion. The minimum angle of 68 ensures no acute angles that could trigger numerical instability.

Convergence Behaviour



(a) Inviscid simulation. **(b)** k-ω simulation.

Figure 1. RMS residual convergence for momentum and mass.

The inviscid simulation converges rapidly: without viscosity, numerical diffusion is minimal and the solver stabilises early, reaching the RMS target of 10^{-6} for all equations. Pressure and velocity fields stabilise with well-defined, oscillation-free shock structures.

The k-ω simulation shows a more gradual convergence trend. Initial fluctuations arise as the solver resolves the developing boundary layer and shock–boundary-layer interaction. Residuals nevertheless decrease monotonically below 10^{-6}, confirming a physically consistent solution. The k-ε simulation (not shown separately) exhibits similar behaviour, converging slightly faster due to lower sensitivity to near-wall resolution.

> **Convergence summary**
>
> All three simulations reach RMS $< 10^{-6}$ for momentum, mass, and turbulence quantities. The inviscid case converges fastest; k-ω requires the most iterations due to the complexity of shock–boundary-layer coupling.

Discussion of the Results

Main Task: Flow Analysis and Simulation Comparison

The dominant feature in all simulations is the oblique shock system generated at each ramp deflection. The first shock at $\alpha = 8$ and the second at $\beta = 22$ produce successive pressure jumps that are clearly captured in both the inviscid and viscous runs.

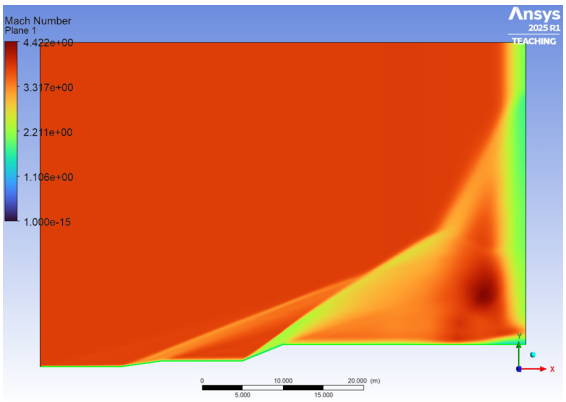


Figure 2. Mach number distribution — k-ω simulation.

In the inviscid case, shock waves are perfectly sharp and no boundary layer develops, so the post-shock Mach number follows theoretical oblique-shock relations closely. In the k-ω simulation, the boundary layer thickens along the lower wall and interacts with the oblique shock, causing a localised pressure footprint upstream of the geometric deflection point and a mild separation bubble near the second ramp corner.

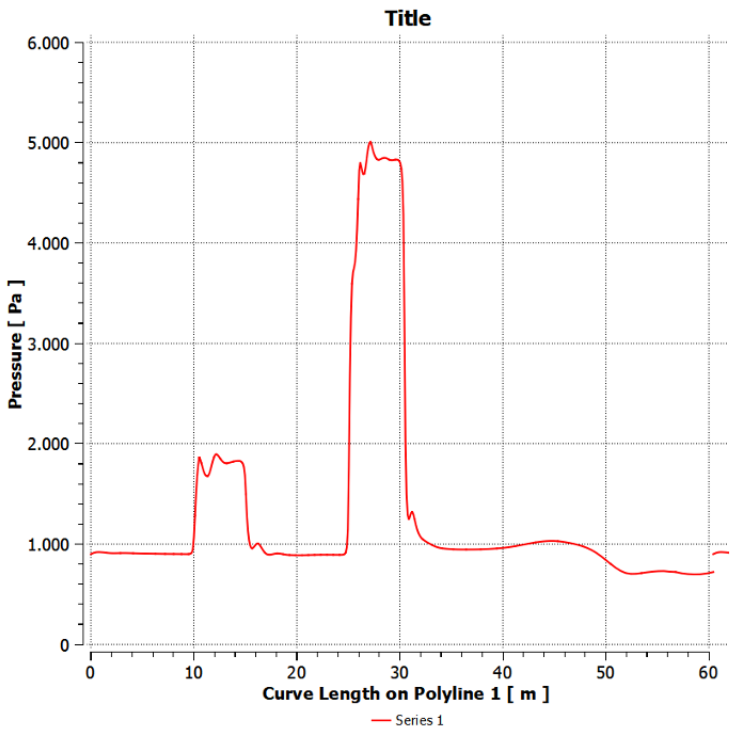


Figure 3. Wall pressure distribution along the ramp for the k-ω simulation.

Key flow quantities

Inlet Mach number: $Ma = 3.8$ Post-shock Mach number (first ramp, inviscid): $Ma_2 \approx 3.1$
Maximum wall pressure (second shock): $\approx 5{,}000$ Pa
Boundary-layer interaction: localised separation observed near the second ramp corner (k-ω).

Special Task: k-ε Model Comparison

The special task requires running the k-ε model and comparing near-wall predictions with the k-ω results, with particular attention to the undisturbed boundary layer and the streamwise velocity profile ahead of the first ramp.

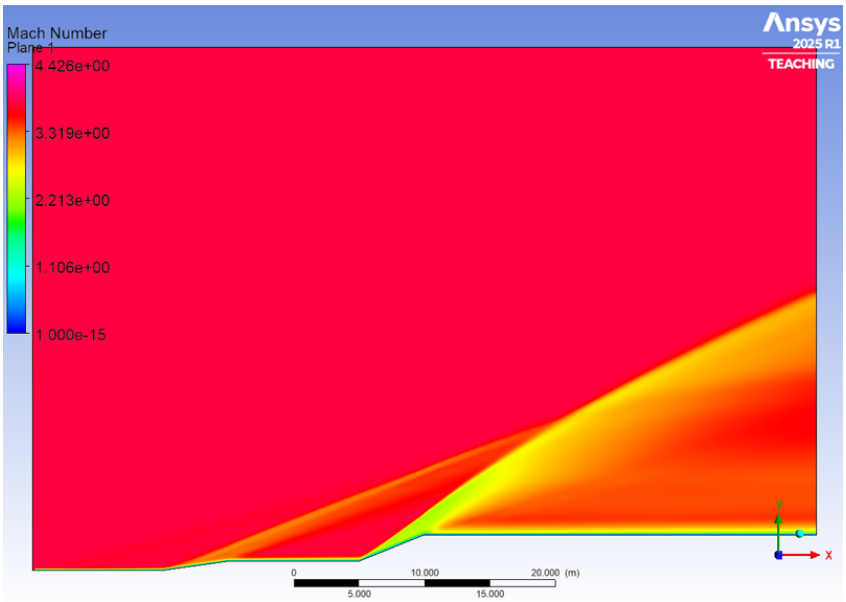


Figure 4. k-ε simulation.

The k-ε model captures the external shock structure well, reproducing the oblique shocks at the correct angles. However, it underpredicts the near-wall velocity gradient in the developing boundary layer ahead of the first ramp. This is consistent with the model's known limitation: the ε-equation becomes ill-conditioned close to solid walls, leading to an overly diffuse velocity profile in the viscous sublayer. The k-ω model, by contrast, is formulated in terms of the specific dissipation rate ω, which is well-defined at the wall, and therefore produces a sharper and more physically accurate boundary-layer profile.

In terms of the outer flow, both models give comparable shock angles and post-shock Mach numbers, confirming that the choice of turbulence model has only a secondary influence on the inviscid core but a primary influence on near-wall quantities such as skin friction and heat transfer.

> **k-ω vs. k-ε — summary**
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> External flow (shock angles, Mach distribution): equivalent results from both models.
> Near-wall boundary layer ahead of the first ramp: k-ω predicts a sharper, physically more accurate profile; k-ε overestimates the boundary-layer thickness.
> Recommended model for this configuration: k-ω due to superior wall treatment under adverse pressure gradients.